

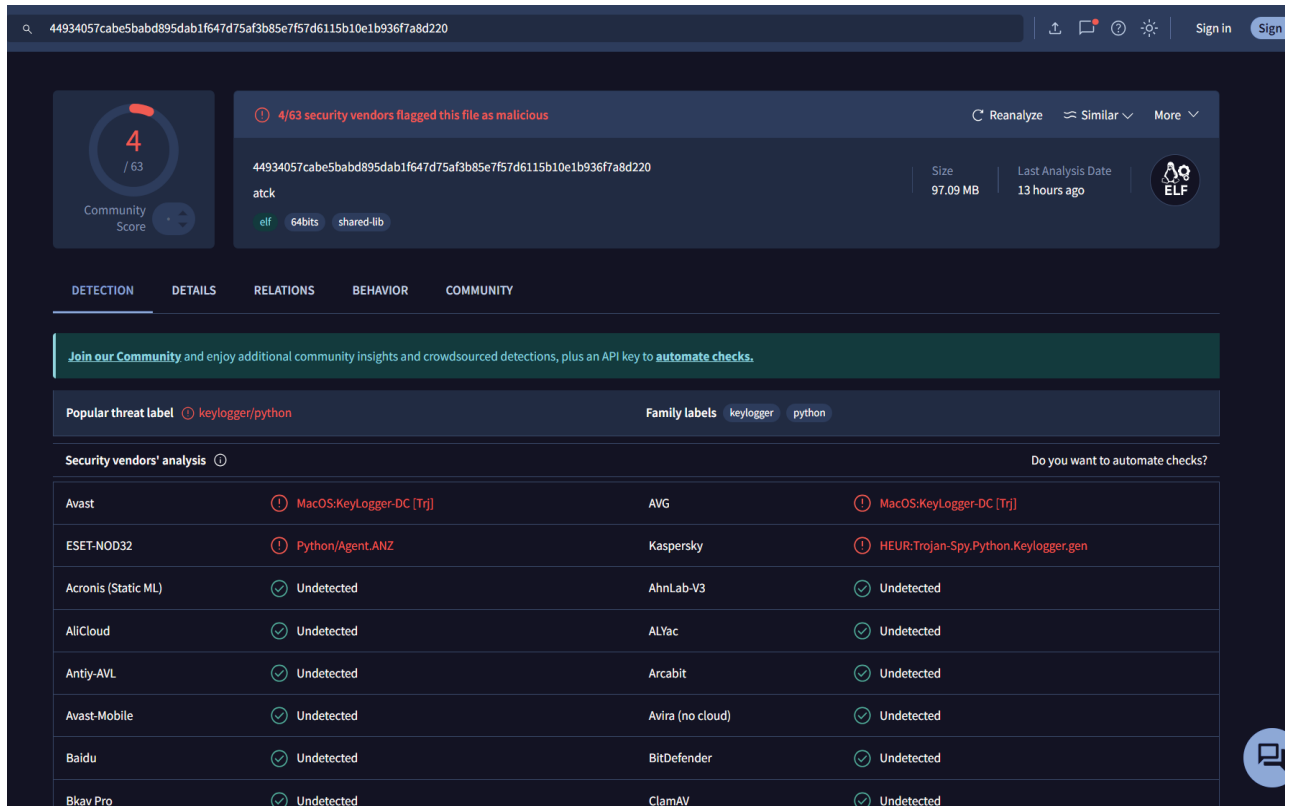


Figure 5: VirusTotal results showing only 4/64 detections

Making it stealthy

Writing a payload is one thing; making it undetectable is another. Once the Python scripts were working, I used 'PyInstaller' to convert the target-side script into a standalone (.elf) binary (make sure you have UPX [Ultimate Packer for eXecutables] installed on your system):

```
pyinstaller --onefile --clean attackscript.py
```

This packed everything into a single Linux executable — clean, portable, and ready to deploy. To avoid static detection:

- I moved imports like 'cv2', 'pynput', and 'sounddevice' inside the functions that used them.
- I renamed suspicious variables and strings.
- I encoded identifiable content using base64 or XOR techniques.
- I used base64-encoded JSON for all communications to slightly mask what was being transferred.

The result: Near-total anti-virus evasion

Once compiled, I submitted the binary to VirusTotal for testing. Out of 64 engines, only four flagged the file. Most notably, well-known, and widely deployed anti-virus engines failed to detect it.

This result was not just surprising, it was alarming. It proved that with custom code, even basic remote access tools can bypass most next-gen antivirus products if they are not behaving in obviously malicious ways or using known exploit frameworks.

Final thoughts

As a cybersecurity specialist, this project was one of the most educational experiments I have done. Writing a custom reverse shell from scratch helped me deeply understand how malware operates, and why anti-virus solutions increasingly rely on behavioural analysis over signature-based detection. It also reinforced a critical lesson: relying solely on traditional anti-virus and EDR solutions is not enough. If you are a red teamer, penetration tester, or simply a curious hacker, building your own tools is one of the best ways to outsmart automated defences and test the cybersecurity of your organisation. **END** 🐧

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